

## SAT Math Mastery · Week 9A

### Solids, Coordinate Geometry & Composite Shapes

|                    |                                                                                                                                                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Time</b>        | 150 minutes: 110 min solving + 40 min self-check and error log                                                                                                                                                         |
| <b>Topics</b>      | volume and surface area of solids · scaling laws · composite, glued and inscribed solids · distance, midpoint, slope · area from coordinates · composite and shaded regions · rate of fill and unit volume · dilations |
| <b>Name / Date</b> | _____                                                                                                                                                                                                                  |

#### 1 · SKILLS & IDEAS

- S1.** Volume of prisms, cylinders, cones, spheres and pyramids; recover a radius from a circumference, a volume or a surface area.
- S2.** Surface area: total vs lateral, slant height of a cone or pyramid, surface area of a prism from its three face pairs.
- S3.** Scaling laws: lengths  $\times k$ , areas  $\times k^2$ , volumes  $\times k^3$ ; recover  $k$  from an area or volume ratio.
- S4.** Composite, glued, removed and inscribed solids (sphere in a cube, cylinder in a sphere, joined prisms hide two faces).
- S5.** Coordinate geometry: distance, midpoint (including a missing endpoint), slope, equidistant points.
- S6.** Area from coordinates: base and height on a grid, decomposition, and the shoelace formula.
- S7.** Composite and shaded plane regions: subtract a part from a whole; equilateral-triangle trapezoids; algebraic side lengths leading to a quadratic.
- S8.** Rate of fill, unit-volume conversions ( $\text{m}^3 \leftrightarrow \text{L}$ ), and dilations in the coordinate plane.

#### 2 · REQUIRED FACTS

- Volume of any prism or cylinder is  $V = Bh$  (area of the base times the height); a cone or pyramid with the same base and height holds exactly  $\frac{1}{3}$  of that.
- The slant height  $\ell$ , the height  $h$  and the base radius  $r$  of a right cone form a right triangle:  $\ell^2 = r^2 + h^2$ .
- A cube's space diagonal is  $s\sqrt{3}$ ; a rectangular box's is  $\sqrt{a^2 + b^2 + c^2}$ . A rectangle inscribed in a circle has its diagonal equal to the diameter.
- When two solids are glued along a face, the surface area is the sum of the two surface areas minus *twice* the area of the glued face.
- Similar figures: if lengths scale by  $k$ , areas scale by  $k^2$  and volumes by  $k^3$ ; angle measures never change.
- $1 \text{ m}^3 = 1000 \text{ L}$  and  $1 \text{ cm}^3 = 1 \text{ mL}$ ; square and cubic units convert with the *square* and *cube* of the length factor.
- A dilation centered at the origin with scale factor  $k$  sends  $(x, y) \rightarrow (kx, ky)$ , multiplies lengths by  $k$  and areas by  $k^2$ .

#### 3 · REQUIRED FORMULAS

$$V_{\text{prism}} = Bh, \quad V_{\text{cyl}} = \pi r^2 h, \quad V_{\text{cone}} = \frac{1}{3}\pi r^2 h, \quad V_{\text{sphere}} = \frac{4}{3}\pi r^3, \quad V_{\text{pyramid}} = \frac{1}{3}Bh$$

$$SA_{\text{sphere}} = 4\pi r^2, \quad SA_{\text{cyl}} = 2\pi r^2 + 2\pi rh, \quad SA_{\text{cone}} = \pi r^2 + \pi r\ell, \quad SA_{\text{box}} = 2(ab + bc + ac)$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}, \quad M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right), \quad \text{Area} = \frac{1}{2} \left| \sum (x_i y_{i+1} - x_{i+1} y_i) \right|$$

#### 4 · MINI THEORY SHEET

Choose the solid's key quantity first.

| Given                             | Fastest route                                                                                               |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------|
| a circumference $C$               | $r = \frac{C}{2\pi}$ , then use $r^2$ in the area or volume                                                 |
| radius and slant height of a cone | $h = \sqrt{\ell^2 - r^2}$ (look for 6-8-10, 5-12-13)                                                        |
| a volume $= k\pi$                 | divide by $\pi$ first; solve $r^3$ or $r^2h$ with whole-number guesses                                      |
| lateral area of a pyramid         | each triangle $= \frac{1}{2}(\text{side})\ell$ ; find $\ell$ , then $h = \sqrt{\ell^2 - (\text{side}/2)^2}$ |
| a rate of fill $Q$ per minute     | height rate $= \frac{Q}{\text{base area}}$ (not divided by the radius)                                      |

**Inscribed solids.** Draw the cross section: a sphere in a cube touches the faces (diameter = edge); a sphere around a cube passes through the vertices (diameter = space diagonal); a cylinder in a sphere has the sphere's radius as the hypotenuse of a right triangle with the half-height and the base radius.

**Scaling.** If the ratio of areas is  $a : b$ , then  $k = \sqrt{b/a}$ ; if the ratio of volumes is  $a : b$ , then  $k = \sqrt[3]{b/a}$ . Convert, then apply to the quantity asked.

**Area from coordinates.** Axis-parallel side  $\rightarrow$  base and height. Otherwise use the shoelace sum, or box the figure and subtract corner triangles.

**Unit volume.** Convert every quantity to one unit *before* dividing; a rate in liters per minute needs a volume in liters.

### Speed shortcuts.

| Situation                                 | Shortcut                                                                |
|-------------------------------------------|-------------------------------------------------------------------------|
| ratio of volumes $a : b$                  | lengths $\sqrt[3]{b/a}$ , areas $\left(\sqrt[3]{b/a}\right)^2$          |
| radius $\times m$ , height $\times n$     | volume $\times m^2n$ (a cylinder or cone)                               |
| cube with surface area $SA$               | face $= SA/6$ , edge $= \sqrt{SA/6}$                                    |
| glued prisms                              | $SA = SA_1 + SA_2 - 2(\text{glued face})$ , or rebuild the combined box |
| missing endpoint                          | $B = 2M - A$ coordinatewise                                             |
| cone or sphere with a $\pi$ in the volume | cancel $\pi$ at once and solve for $r$                                  |
| solid of revolution of a right triangle   | cone volume $\frac{1}{3}\pi r^2h$ with $r$ the distance to the axis     |
| dilation by $k$ about the origin          | multiply every coordinate by $k$ ; area $\times k^2$                    |

## 5 · GUIDED EXAMPLES

### Example 1 — Composite solid: cylinder topped by a hemisphere

A silo is a cylinder of radius 3 and height 10 with a hemisphere of radius 3 on top. Find its volume.

Cylinder:  $\pi r^2h = \pi(9)(10) = 90\pi$ . Hemisphere:  $\frac{1}{2} \cdot \frac{4}{3}\pi r^3 = \frac{2}{3}\pi(27) = 18\pi$ .

Total volume  $= 90\pi + 18\pi = 108\pi$ .

### Example 2 — Scaling: area ratio to volume

Two similar prisms have surface areas 50 and 162. The smaller has volume 125. Find the volume of the larger.

Area ratio  $\frac{162}{50} = \frac{81}{25} = k^2$ , so  $k = \frac{9}{5}$ . Volume ratio  $k^3 = \frac{729}{125}$ .

Larger volume  $= 125 \cdot \frac{729}{125} = 729$ .

### Example 3 — Area from coordinates (shoelace)

Find the area of the triangle with vertices  $P(-1, 2)$ ,  $Q(5, -2)$ ,  $R(3, 6)$ .

$\sum (x_i y_{i+1} - x_{i+1} y_i)$ :  $P \rightarrow Q$ :  $(-1)(-2) - (5)(2) = -8$ ;  $Q \rightarrow R$ :  $(5)(6) - (3)(-2) = 36$ ;  $R \rightarrow P$ :  $(3)(2) - (-1)(6) = 12$ .

Sum  $= -8 + 36 + 12 = 40$ , so area  $= \frac{1}{2}|40| = 20$ .

## 6 · PRACTICE SET — 44 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

A right circular cylinder has a height of 5 and a base with circumference  $8\pi$ . What is the volume of the cylinder?

(A)  $40\pi$

(B)  $80\pi$  ✓

(C)  $160\pi$

(D)  $320\pi$

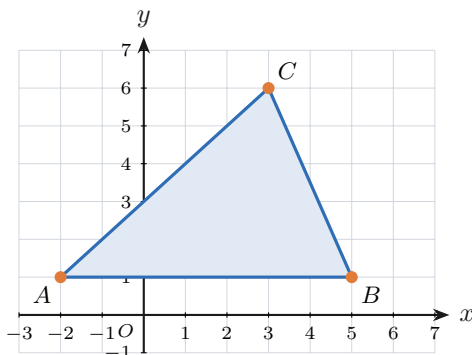
1.  $C = 2\pi r = 8\pi$ , so  $r = 4$ .

2.  $V = \pi r^2 h = \pi(4)^2(5) = 80\pi$ .

Answer: (B)

2. EASY · NO CALCULATOR NEEDED

In the  $xy$ -plane, the vertices of triangle  $ABC$  are plotted as shown. What is the area of triangle  $ABC$ ?



1. Read the vertices:  $A(-2, 1)$ ,  $B(5, 1)$ ,  $C(3, 6)$ .

2.  $\overline{AB}$  is horizontal, so the base is  $5 - (-2) = 7$ .

3. The height is the vertical distance from  $C$  to line  $AB$ :  $6 - 1 = 5$ .

4. Area =  $\frac{1}{2}(7)(5) = 17.5$ .

Answer: 17.5

3. EASY · NO CALCULATOR NEEDED

Point  $M(3, 5)$  is the midpoint of  $\overline{AB}$ . If  $A$  has coordinates  $(-1, 2)$ , what are the coordinates of  $B$ ?

(A)  $\left(1, \frac{7}{2}\right)$

(B)  $(8, 7)$

(C)  $(7, 8)$  ✓

(D)  $(4, 3)$

1. Midpoint formula:  $3 = \frac{-1 + x_B}{2}$  and  $5 = \frac{2 + y_B}{2}$ .

2.  $x_B = 6 + 1 = 7$  and  $y_B = 10 - 2 = 8$ .

3.  $B = (7, 8)$ .

Answer: (C)

4.

EASY · CALCULATOR

A cone has a volume of 14 cubic centimeters. A second cone is similar to the first, and each linear dimension of the second cone is 3 times the corresponding dimension of the first. What is the volume, in cubic centimeters, of the second cone?

- Lengths scale by  $k = 3$ , so volumes scale by  $k^3 = 27$ .
- $14 \cdot 27 = 378$ .

**Answer:** 378

5.

EASY · NO CALCULATOR NEEDED

A cube has a total surface area of 150 square inches. What is the volume of the cube, in cubic inches?

(A) 125 ✓

(B) 150

(C) 625

(D) 25

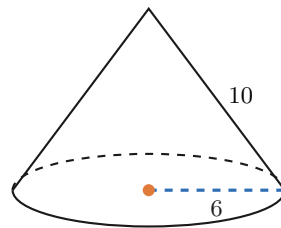
- A cube has 6 identical square faces: each face has area  $\frac{150}{6} = 25$ .
- Edge length  $= \sqrt{25} = 5$ .
- $V = 5^3 = 125$ .

**Answer:** (A)

6.

MEDIUM · CALCULATOR

The cone shown has a base radius of 6 and a slant height of 10. What is the volume of the cone?

(A)  $120\pi$ (B)  $288\pi$ (C)  $360\pi$ (D)  $96\pi$  ✓

- The height, the radius and the slant height form a right triangle:  $h = \sqrt{10^2 - 6^2} = \sqrt{64} = 8$ .
- $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(8) = 96\pi$ .

**Answer:** (D)

7.

MEDIUM · NO CALCULATOR NEEDED

A sphere has a volume of  $288\pi$  cubic centimeters. The surface area of the sphere is  $k\pi$  square centimeters. What is the value of  $k$ ?

1.  $\frac{4}{3}\pi r^3 = 288\pi \Rightarrow r^3 = 216 \Rightarrow r = 6$ .
2.  $SA = 4\pi r^2 = 4\pi(36) = 144\pi$ .
3.  $k = 144$ .

**Answer:** 144

8.

MEDIUM · NO CALCULATOR NEEDED

A cylinder has radius  $r$  and height  $h$ . Which of the following changes to the cylinder results in a volume that is exactly twice the original volume?

- I. Multiply the radius by  $\sqrt{2}$  and leave the height unchanged.
- II. Multiply the radius by 2 and divide the height by 2.
- III. Multiply the radius by 2 and divide the height by 4.

(A) I only

(B) II only

**(C) I and II only ✓**

(D) I, II, and III

1.  $V = \pi r^2 h$ : the volume is multiplied by (radius factor)<sup>2</sup> · (height factor).
2. Statement I:  $(\sqrt{2})^2 \cdot 1 = 2$ , so the volume doubles.
3. Statement II:  $2^2 \cdot \frac{1}{2} = 2$ , so the volume doubles.
4. Statement III:  $2^2 \cdot \frac{1}{4} = 1$ , so the volume is unchanged.
5. Only I and II double the volume.

**Answer:** (C)

9.

MEDIUM · NO CALCULATOR NEEDED

A rectangular box has a square base with side length 6 and a height of  $h$ . The total surface area of the box is 240 square units. What is the value of  $h$ ?

1. Surface area =  $2(\text{base}) + 4(\text{side face}) = 2(6)(6) + 4(6h) = 72 + 24h$ .
2.  $72 + 24h = 240 \Rightarrow 24h = 168$ .
3.  $h = 7$ .

**Answer:** 7

10.

MEDIUM · NO CALCULATOR NEEDED

The endpoints of a diameter of a circle in the  $xy$ -plane are  $(-3, 2)$  and  $(5, 8)$ . What is the radius of the circle?

(A) 100

(B) 5 ✓

(C) 10

(D) 25

1. Diameter  $= \sqrt{(5 - (-3))^2 + (8 - 2)^2} = \sqrt{64 + 36} = 10$ .

2. Radius  $= \frac{10}{2} = 5$ .

**Answer: (B)**

11.

MEDIUM · NO CALCULATOR NEEDED

Triangle  $ABC$  is similar to triangle  $DEF$ . The area of triangle  $ABC$  is 48 and the area of triangle  $DEF$  is 108. The longest side of triangle  $ABC$  has length 10. What is the length of the longest side of triangle  $DEF$ ?

1. Area ratio  $= \frac{108}{48} = \frac{9}{4} = k^2$ , so  $k = \frac{3}{2}$ .

2. Longest side of  $DEF = 10 \cdot \frac{3}{2} = 15$ .

**Answer: 15**

12.

MEDIUM · CALCULATOR

A cylindrical tank has an inside radius of 2 meters. Water flows into the tank at a constant rate of 0.5 cubic meter per minute. Which of the following is closest to the rate, in meters per minute, at which the water level in the tank rises?

(A) 0.08

(B) 0.16

(C) 0.25

(D) 0.04 ✓

1. In one minute the water added is  $0.5 \text{ m}^3$ , spread over the circular base of area  $\pi r^2 = 4\pi \text{ m}^2$ .

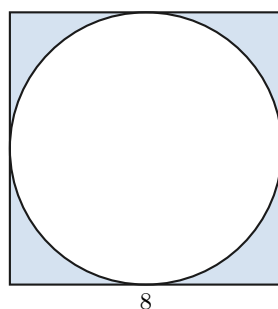
2. Rate of rise  $= \frac{0.5}{4\pi} = \frac{1}{8\pi} \approx 0.040$  meters per minute.

**Answer: (D)**

13.

MEDIUM · NO CALCULATOR NEEDED

A circle is inscribed in a square with side length 8, as shown. What is the area of the shaded region?

**(A)**  $64 - 16\pi$  ✓(B)  $64 - 8\pi$ (C)  $64 - 4\pi$ (D)  $16\pi$ 

1. The circle touches all four sides, so its diameter is 8 and its radius is 4.
2. Circle area  $= \pi(4)^2 = 16\pi$ ; square area  $= 64$ .
3. Shaded area  $= 64 - 16\pi$ .

**Answer: (A)**

14.

HARD · CALCULATOR

A solid sphere with a radius of 3 centimeters is placed in a cylindrical container that holds water and has an inside base radius of 5 centimeters. The sphere is completely submerged and no water spills. By how many centimeters does the water level rise?

1. The rise in water equals the volume of the sphere:  $\frac{4}{3}\pi(3)^3 = 36\pi \text{ cm}^3$ .
2. The water level rises over the base area  $\pi(5)^2 = 25\pi \text{ cm}^2$ .
3. Rise  $= \frac{36\pi}{25\pi} = \frac{36}{25} = 1.44 \text{ cm}$ .

**Answer: 1.44**

15.

HARD · NO CALCULATOR NEEDED

All eight vertices of a cube with edge length 6 lie on the surface of a sphere. What is the volume of the sphere?

(A)  $72\sqrt{2}\pi$

**(B)  $108\sqrt{3}\pi$  ✓**

(C)  $288\pi$

(D)  $36\pi$

1. The sphere's diameter is the cube's space diagonal:  $6\sqrt{3}$ , so  $r = 3\sqrt{3}$ .

2.  $r^3 = (3\sqrt{3})^3 = 27 \cdot 3\sqrt{3} = 81\sqrt{3}$ .

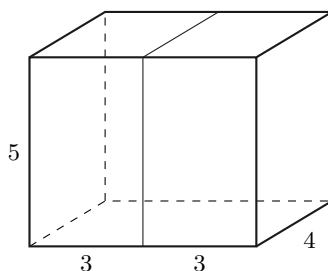
3.  $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi \cdot 81\sqrt{3} = 108\sqrt{3}\pi$ .

**Answer: (B)**

16.

HARD · NO CALCULATOR NEEDED

Two identical rectangular blocks, each measuring 3 by 4 by 5, are glued together along a 4-by-5 face to form one larger rectangular block, as shown. What is the total surface area of the larger block?



(A) 188

(B) 94

**(C) 148 ✓**

(D) 168

1. One block:  $2(3 \cdot 4 + 3 \cdot 5 + 4 \cdot 5) = 2(47) = 94$ .

2. Two blocks:  $2 \cdot 94 = 188$ . The glued faces are hidden: subtract  $2 \cdot (4 \cdot 5) = 40$ .

3.  $188 - 40 = 148$ . (Check: the new block is  $6 \times 4 \times 5$ , with  $2(24 + 30 + 20) = 148$ .)

**Answer: (C)**

17.

HARD · CALCULATOR

The four lateral faces of a right square pyramid are congruent triangles with a combined area of 260. The base of the pyramid is a square with side length 10. What is the volume of the pyramid?

(A) 400 ✓

(B) 1200

(C) 1300

(D)  $\frac{1300}{3}$ 

1. Each lateral face has area  $\frac{260}{4} = 65 = \frac{1}{2}(10)\ell$ , so the slant height is  $\ell = 13$ .

2. The height, half the base side (5) and the slant height form a right triangle:  $h = \sqrt{13^2 - 5^2} = 12$ .

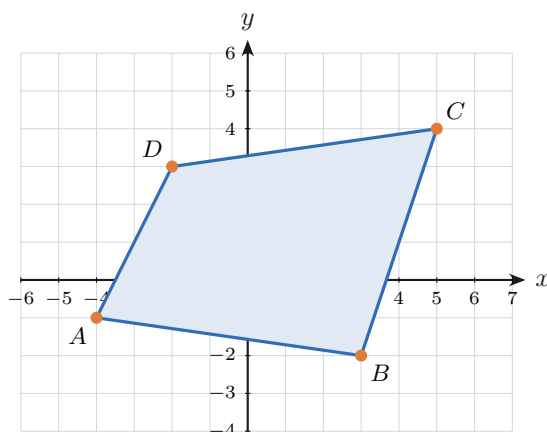
3.  $V = \frac{1}{3}Bh = \frac{1}{3}(100)(12) = 400$ .

Answer: (A)

18.

HARD · CALCULATOR

In the  $xy$ -plane, quadrilateral  $ABCD$  has the vertices shown on the grid. What is the area of quadrilateral  $ABCD$ ?



(A) 70

(B)  $\frac{63}{2}$ 

(C) 38

(D) 35 ✓

1. Read the vertices in order:  $A(-4, -1)$ ,  $B(3, -2)$ ,  $C(5, 4)$ ,  $D(-2, 3)$ .

2.  $A \rightarrow B$ :  $(-4)(-2) - (3)(-1) = 11$ .  $B \rightarrow C$ :  $(3)(4) - (5)(-2) = 22$ .

3.  $C \rightarrow D$ :  $(5)(3) - (-2)(4) = 23$ .  $D \rightarrow A$ :  $(-2)(-1) - (-4)(3) = 14$ .

4. Sum =  $11 + 22 + 23 + 14 = 70$ , so the area is  $\frac{1}{2}(70) = 35$ .

Answer: (D)

**19.**

HARD · NO CALCULATOR NEEDED

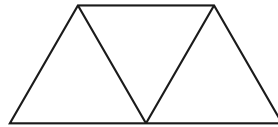
A rectangle with side lengths 10 and 24 is inscribed in a circle. The area of the circle is  $k\pi$ . What is the value of  $k$ ?

1. The diagonal of an inscribed rectangle is a diameter:  $d = \sqrt{10^2 + 24^2} = \sqrt{676} = 26$ .
2.  $r = 13$ , so the area is  $\pi(13)^2 = 169\pi$ .
3.  $k = 169$ .

**Answer:** 169**20.**

HARD · CALCULATOR

The trapezoid shown is composed of three congruent equilateral triangles. The perimeter of the trapezoid is 20. What is the area of the trapezoid?

(A)  $48\sqrt{3}$ (B)  $4\sqrt{3}$ **(C)  $12\sqrt{3}$  ✓**(D)  $16\sqrt{3}$ 

1. The perimeter consists of 5 sides each equal to the triangle side  $s$ :  $5s = 20 \Rightarrow s = 4$ .
2. Area of one equilateral triangle  $= \frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4}(16) = 4\sqrt{3}$ .
3. Three triangles:  $3 \cdot 4\sqrt{3} = 12\sqrt{3}$ .

**Answer:** (C)

**21.**

HARD · CALCULATOR

A rectangular pool measures 6 meters by 4 meters and is 1.5 meters deep. A pump fills the empty pool at a constant rate of 150 liters per minute. How many hours does it take to fill the pool completely? (1 cubic meter = 1000 liters)

1. Volume =  $6 \cdot 4 \cdot 1.5 = 36 \text{ m}^3 = 36,000$  liters.

2. Time =  $\frac{36,000}{150} = 240$  minutes.

3.  $240 \div 60 = 4$  hours.

**Answer:** 4

**22.**

HARD · NO CALCULATOR NEEDED

Two solid cones are similar. The volume of the larger cone is  $\frac{27}{8}$  times the volume of the smaller cone. The surface area of the smaller cone is 100 square centimeters. What is the surface area, in square centimeters, of the larger cone?

(A) 150

**(B) 225 ✓**

(C) 337.5

(D) 800

1. Volume ratio  $k^3 = \frac{27}{8}$ , so the length ratio is  $k = \frac{3}{2}$ .

2. Surface areas scale by  $k^2 = \frac{9}{4}$ .

3.  $100 \cdot \frac{9}{4} = 225$ .

**Answer:** (B)

**23.**

HARD · CALCULATOR

In the  $xy$ -plane, triangle  $ABC$  has vertices  $A(1, 2)$ ,  $B(7, 2)$  and  $C(1, 10)$ . Triangle  $DEF$  is the image of triangle  $ABC$  under a dilation centered at the origin, where  $D$  is the image of  $A$  and  $D = (3, 6)$ . What is the area of triangle  $DEF$ ?

(A) 648

(B) 72

(C) 144

(D) 216 ✓

1.  $A(1, 2) \rightarrow D(3, 6)$ : each coordinate is multiplied by 3, so  $k = 3$ .

2. Area of  $ABC$ : the base  $\overline{AB} = 6$  is horizontal and the height  $\overline{AC} = 8$  is vertical, so area  $= \frac{1}{2}(6)(8) = 24$ .

3. Areas scale by  $k^2 = 9$ :  $24 \cdot 9 = 216$ .

**Answer: (D)**

**24.**

SAT CHALLENGE · CALCULATOR

A cone-shaped paper cup is held point down. The cup has an inside radius of 6 centimeters at the top and a height of 12 centimeters. The cup contains water to a depth of 8 centimeters. All of this water is poured into an empty cylindrical glass with an inside radius of 4 centimeters. What is the depth, in centimeters, of the water in the glass?

1. The water forms a smaller cone similar to the cup, with height 8: its radius is  $6 \cdot \frac{8}{12} = 4$ .

2. Water volume  $= \frac{1}{3}\pi(4)^2(8) = \frac{128\pi}{3}$ .

3. In the glass,  $\pi(4)^2d = \frac{128\pi}{3} \Rightarrow 16d = \frac{128}{3}$ .

4.  $d = \frac{8}{3} \approx 2.67$ .

**Answer:  $\frac{8}{3}$**

**25.**

SAT CHALLENGE · DESMOS

In the  $xy$ -plane, point  $P$  lies on the line  $y = 2x + 1$  and is the same distance from  $A(-1, 6)$  as from  $B(5, 0)$ . What is the area of triangle  $ABP$ ?

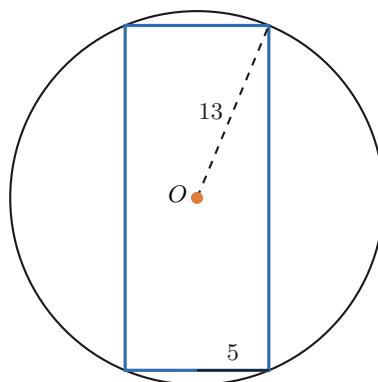
1.  $P$  lies on the perpendicular bisector of  $\overline{AB}$ . Midpoint of  $\overline{AB}$ :  $(2, 3)$ ; slope of  $\overline{AB} = \frac{0 - 6}{5 - (-1)} = -1$ , so the bisector has slope 1:  $y = x + 1$ .
2. Intersect with  $y = 2x + 1$ :  $x + 1 = 2x + 1 \Rightarrow x = 0$ , so  $P = (0, 1)$ .
3. Shoelace with  $A(-1, 6)$ ,  $B(5, 0)$ ,  $P(0, 1)$ :  $(-1)(0) - (5)(6) = -30$ ;  $(5)(1) - (0)(0) = 5$ ;  $(0)(6) - (-1)(1) = 1$ . Sum =  $-24$ .
4. Area =  $\frac{1}{2}|-24| = 12$ .

**Answer:** 12

26.

SAT CHALLENGE · CALCULATOR

A right circular cylinder is inscribed in a sphere with radius 13 so that the circumference of each base of the cylinder lies on the sphere. The base of the cylinder has radius 5. The figure shows a cross section through the center  $O$  of the sphere. What is the volume of the cylinder?

**(A)**  $600\pi$  ✓(B)  $650\pi$ (C)  $300\pi$ (D)  $325\pi$ 

1. In the cross section, a radius of the sphere to a corner of the rectangle is the hypotenuse of a right triangle with legs 5 (the base radius) and half the cylinder's height.
2. Half-height =  $\sqrt{13^2 - 5^2} = \sqrt{144} = 12$ , so the height is 24.
3.  $V = \pi r^2 h = \pi(25)(24) = 600\pi$ .

**Answer: (A)**

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

A scale model of a water tank holds 8 liters of water. The actual tank holds 27 cubic meters of water, and the model is geometrically similar to the actual tank. The model is 36 centimeters tall. How tall, in meters, is the actual tank? (1 cubic meter = 1000 liters)

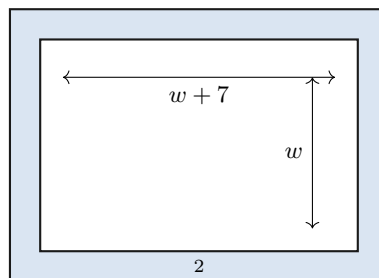
1. Actual capacity:  $27 \text{ m}^3 = 27,000$  liters.
2. Volume ratio =  $\frac{27,000}{8} = 3375 = k^3$ , so  $k = 15$ .
3. Actual height =  $36 \cdot 15 = 540 \text{ cm} = 5.4 \text{ m}$ .

**Answer:** 5.4

28.

SAT CHALLENGE · CALCULATOR

A rectangular garden has a width of  $w$  meters and a length that is 7 meters greater than its width. A path of uniform width 2 meters surrounds the garden on all sides. The area of the path alone is 156 square meters. What is the area, in square meters, of the garden?



(A) 98

**(B) 294 ✓**

(C) 450

(D) 588

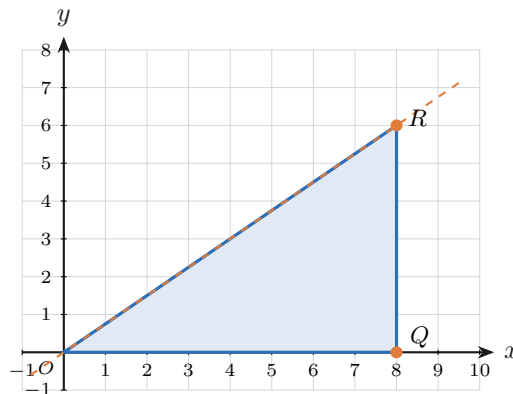
1. The outer rectangle is  $(w + 4)$  by  $(w + 11)$  because the path adds 2 meters on each side.
2. Path area =  $(w + 4)(w + 11) - w(w + 7) = (w^2 + 15w + 44) - (w^2 + 7w) = 8w + 44$ .
3.  $8w + 44 = 156 \Rightarrow w = 14$ .
4. Garden area =  $w(w + 7) = 14 \cdot 21 = 294$ .

**Answer:** (B)

29.

SAT CHALLENGE · CALCULATOR

In the  $xy$ -plane, triangle  $OQR$  has vertices  $O(0,0)$ ,  $Q(8,0)$  and  $R(8,6)$ , as shown. The triangle is rotated  $360^\circ$  about the line containing side  $\overline{OR}$  to form a solid. What is the volume of the solid?



(A)  $128\pi$

(B)  $\frac{1152}{5}\pi$

(C)  $\frac{384}{5}\pi$  ✓

(D)  $96\pi$

1.  $OR = \sqrt{8^2 + 6^2} = 10$ . The solid is two cones sharing a base circle whose radius is the distance from  $Q$  to line  $OR$ .

2. That radius is the altitude to the hypotenuse:  $\frac{8 \cdot 6}{10} = 4.8$ .

3. The two cones have heights that add to  $OR = 10$ , so  $V = \frac{1}{3}\pi(4.8)^2(10)$ .

4.  $V = \frac{1}{3}\pi(23.04)(10) = 76.8\pi = \frac{384}{5}\pi$ .

Answer: (C)

**30.**

SAT CHALLENGE · CALCULATOR

A solid metal sphere with radius 9 centimeters is melted. Exactly 90% of the metal is recast into identical solid cones, each with radius 3 centimeters and height 4 centimeters. What is the greatest number of complete cones that can be made?

1. Sphere volume =  $\frac{4}{3}\pi(9)^3 = 972\pi$ .
2. Metal used:  $0.9 \cdot 972\pi = 874.8\pi$ .
3. One cone:  $\frac{1}{3}\pi(3)^2(4) = 12\pi$ .
4.  $\frac{874.8\pi}{12\pi} = 72.9$ , so at most 72 complete cones can be made.

**Answer:** 72**31.**

SAT CHALLENGE · NO CALCULATOR NEEDED

Two similar rectangular crates have surface areas of 144 square feet and 400 square feet. The smaller crate has a volume of 108 cubic feet. What is the volume, in cubic feet, of the larger crate?

**(A)** 500 ✓

(B) 1500

(C) 180

(D) 300

1. Areas scale by  $k^2$ :  $k^2 = \frac{400}{144} = \frac{25}{9}$ , so the linear scale factor is  $k = \frac{5}{3}$ .
2. Volumes scale by  $k^3 = \frac{125}{27}$ .
3.  $108 \cdot \frac{125}{27} = 4 \cdot 125 = 500$ .
4. (Check with dimensions  $3 \times 6 \times 6$ : surface area 144, volume 108; scaled by  $\frac{5}{3}$ :  $5 \times 10 \times 10$ , surface area 400, volume 500.)

**Answer:** (A)

**32.**

SAT CHALLENGE · NO CALCULATOR NEEDED

Two similar solid blocks have surface areas of 208 square centimeters and 325 square centimeters. The larger block has a volume of 375 cubic centimeters. What is the volume, in cubic centimeters, of the smaller block?

1. Going from the larger to the smaller block, areas scale by  $\frac{208}{325} = \frac{16}{25}$ .

2. So the linear factor is  $\sqrt{\frac{16}{25}} = \frac{4}{5}$ .

3. Volume scales by  $\left(\frac{4}{5}\right)^3 = \frac{64}{125}$ .

4.  $375 \cdot \frac{64}{125} = 3 \cdot 64 = 192$ .

**Answer:** 192

**33.**

SAT CHALLENGE · CALCULATOR

A closed right circular cylinder has a volume of  $48\pi$  cubic centimeters and a total surface area of  $56\pi$  square centimeters. The height of the cylinder is greater than its diameter. What is the height, in centimeters, of the cylinder?

(A) 2

(B) 3

(C) 4

**(D) 12 ✓**

1. From  $\pi r^2 h = 48\pi$ :  $h = \frac{48}{r^2}$ .

2. Surface area:  $2\pi r^2 + 2\pi r h = 56\pi \Rightarrow r^2 + r h = 28 \Rightarrow r^2 + \frac{48}{r} = 28$ .

3. Multiply by  $r$ :  $r^3 - 28r + 48 = 0$ . Test divisors of 48:  $r = 2$  gives  $8 - 56 + 48 = 0$ ,  $r = 4$  gives  $64 - 112 + 48 = 0$ . Factor:  $(r - 2)(r - 4)(r + 6) = 0$ , so the positive radii are 2 and 4.

4.  $r = 2$ :  $h = 12 > 4 = 2r$ .  $r = 4$ :  $h = 3 < 8 = 2r$ , which violates the condition.

5. The height is 12 centimeters. (The other choices are the radii 2 and 4 and the height 3 of the rejected solution.)

**Answer:** (D)

34.

SAT CHALLENGE · NO CALCULATOR NEEDED

Sphere  $A$  has a great circle with circumference  $8\pi$  centimeters. The surface area of sphere  $B$  is  $80\pi$  square centimeters greater than the surface area of sphere  $A$ . What is the ratio of the volume of sphere  $B$  to the volume of sphere  $A$ ?

- (A)  $\frac{3}{2}$  (B)  $\frac{9}{4}$   
 (C)  $\frac{27}{8}$  ✓ (D)  $\frac{5}{4}$

- A great circle of sphere  $A$  has circumference  $2\pi r_A = 8\pi$ , so  $r_A = 4$  and its surface area is  $4\pi(16) = 64\pi$ .
- Sphere  $B$ :  $4\pi r_B^2 = 64\pi + 80\pi = 144\pi \Rightarrow r_B^2 = 36 \Rightarrow r_B = 6$ .
- Radius ratio  $\frac{6}{4} = \frac{3}{2}$ . Volumes scale with the cube:  $\left(\frac{3}{2}\right)^3 = \frac{27}{8}$ .
- (Other choices:  $\frac{3}{2}$  is the radius ratio,  $\frac{9}{4}$  is the area ratio, and  $\frac{5}{4} = \frac{80}{64}$  compares the area difference with sphere  $A$ 's area.)

**Answer: (C)**

35.

SAT CHALLENGE · CALCULATOR

A right pyramid has a square base. The diagonal of the base is  $10\sqrt{2}$  centimeters and the volume of the pyramid is 400 cubic centimeters. What is the lateral surface area, in square centimeters, of the pyramid (the four triangular faces, not the base)?

- (A) 360 (B) 260 ✓  
 (C) 240 (D) 130

- The base diagonal is  $s\sqrt{2} = 10\sqrt{2}$ , so the side is  $s = 10$  and the base area is 100.
- Volume  $\frac{1}{3}(100)h = 400 \Rightarrow h = 12$ .
- The slant height (apex to the midpoint of a base edge) is the hypotenuse of a right triangle with legs  $h = 12$  and  $\frac{s}{2} = 5$ :  
 $\ell = \sqrt{144 + 25} = 13$ .
- Each face has area  $\frac{1}{2}(10)(13) = 65$ , so the lateral area is  $4 \cdot 65 = 260$ .
- (Using the height 12 instead of 13 gives 240; two faces give 130; adding the base gives 360.)

**Answer: (B)**

**36.**

SAT CHALLENGE · CALCULATOR

A metal pipe is a hollow cylinder of length 15 centimeters and outer radius 10 centimeters. The wall has a uniform thickness of  $t$  centimeters, where  $0 < t < 10$ . The volume of metal in the pipe is  $765\pi$  cubic centimeters. What is the value of  $t$ ?

1. The inner radius is  $10 - t$ . Metal volume  $= \pi(15) [10^2 - (10 - t)^2] = 765\pi$ .

2.  $100 - (10 - t)^2 = \frac{765}{15} = 51 \Rightarrow (10 - t)^2 = 49$ .

3.  $10 - t = 7$  or  $10 - t = -7$ , so  $t = 3$  or  $t = 17$ .

4. Only  $t = 3$  satisfies  $0 < t < 10$ .

**Answer: 3****37.**

SAT CHALLENGE · NO CALCULATOR NEEDED

A right pyramid has a rectangular base. Its height is 12 centimeters, and the slant heights of its triangular faces are 13 centimeters for one pair of opposite faces and 20 centimeters for the other pair. What is the total surface area, in square centimeters, of the pyramid, including the base?

(A) 1160

(B) 616

(C) 866

**(D) 936 ✓**

1. The slant height of a face is the hypotenuse of a right triangle with legs 12 and half the base side that face sits on.

2. Slant height 13: half-side  $= \sqrt{13^2 - 12^2} = 5$ , so that face sits on a side of length 10. Slant height 20: half-side  $= \sqrt{20^2 - 12^2} = 16$ , so that face sits on a side of length 32.

3. Base area:  $10 \cdot 32 = 320$ .

4. Two faces with base 32 and slant height 13:  $2 \cdot \frac{1}{2} \cdot 32 \cdot 13 = 416$ . Two faces with base 10 and slant height 20:  $2 \cdot \frac{1}{2} \cdot 10 \cdot 20 = 200$ .

5. Total:  $320 + 416 + 200 = 936$ .

**Answer: (D)**

**38.**

SAT CHALLENGE · CALCULATOR

A raised garden bed is filled with soil. A vertical cross-section of the soil, perpendicular to the 12-foot length of the bed, is a trapezoid: the bed is 8 feet wide, and the soil is 1.5 feet deep along one side and 3.5 feet deep along the opposite side, with the surface sloping uniformly between them. Soil is sold at \$45 per cubic yard and billed for the exact volume (1 yard = 3 feet). What is the cost, in dollars, of the soil needed?

1. Trapezoid cross-section: parallel sides (depths) 1.5 and 3.5 ft, height (width) 8 ft. Area =  $\frac{1.5 + 3.5}{2} \cdot 8 = 20 \text{ ft}^2$ .
2. Volume =  $20 \cdot 12 = 240 \text{ ft}^3$ .
3.  $1 \text{ yd}^3 = 3^3 = 27 \text{ ft}^3$ , so the volume is  $\frac{240}{27} = \frac{80}{9} \text{ yd}^3$ .
4. Cost =  $\frac{80}{9} \cdot 45 = 80 \cdot 5 = 400$  dollars.

**Answer:** 400**39.**

SAT CHALLENGE · CALCULATOR

Cylinder  $P$  has radius 2 centimeters and height 6 centimeters. Cylinder  $Q$  is similar to cylinder  $P$  and has a total surface area of  $200\pi$  square centimeters. What is the volume, in cubic centimeters, of cylinder  $Q$ ?

**(A)**  $375\pi$  ✓(B)  $600\pi$ (C)  $60\pi$ (D)  $150\pi$ 

1. Total surface area of  $P$ :  $2\pi r(r + h) = 2\pi(2)(8) = 32\pi$ . Volume of  $P$ :  $\pi r^2 h = 24\pi$ .
2. Area ratio =  $\frac{200\pi}{32\pi} = \frac{25}{4}$ , so the linear factor is  $k = \sqrt{\frac{25}{4}} = \frac{5}{2}$ .
3. Volume ratio =  $k^3 = \frac{125}{8}$ .
4.  $V_Q = 24\pi \cdot \frac{125}{8} = 375\pi$ .
5. (Check:  $Q$  has radius 5, height 15; surface area  $2\pi(5)(20) = 200\pi$ , volume  $\pi(25)(15) = 375\pi$ .)

**Answer:** (A)

40.

SAT CHALLENGE · CALCULATOR

Rectangular box  $P$  has dimensions 3 inches by 4 inches by 6 inches. Box  $Q$  is similar to box  $P$  and has a volume of 1125 cubic inches. What is the total surface area, in square inches, of box  $Q$ ?

1. Volume of  $P$ :  $3 \cdot 4 \cdot 6 = 72$ . Surface area of  $P$ :  $2(12 + 18 + 24) = 108$ .

2. Volume ratio  $\frac{1125}{72} = \frac{125}{8}$ , so the linear factor is  $k = \sqrt[3]{\frac{125}{8}} = \frac{5}{2}$ .

3. Areas scale by  $k^2 = \frac{25}{4}$ :  $108 \cdot \frac{25}{4} = 675$ .

4. (Check:  $Q$  is  $7.5 \times 10 \times 15$ , volume 1125, surface area  $2(75 + 112.5 + 150) = 675$ .)

**Answer:** 675

41.

SAT CHALLENGE · NO CALCULATOR NEEDED

A sphere is inscribed in a right circular cone whose base has radius 5 and whose height is 12, so that the sphere is tangent to the base and to the lateral surface of the cone. What fraction of the volume of the cone is occupied by the sphere? (Enter your answer as a fraction.)

1. Slice through the axis: the cross-section is an isosceles triangle with base 10, height 12 and slant sides  $\sqrt{5^2 + 12^2} = 13$ . The sphere appears as the circle inscribed in this triangle.

2. Area =  $\frac{1}{2}(10)(12) = 60$ . Semiperimeter  $s = \frac{13 + 13 + 10}{2} = 18$ . Inradius  $r = \frac{60}{18} = \frac{10}{3}$ .

3. Sphere volume:  $\frac{4}{3}\pi \left(\frac{10}{3}\right)^3 = \frac{4000\pi}{81}$ . Cone volume:  $\frac{1}{3}\pi(5^2)(12) = 100\pi$ .

4. Fraction =  $\frac{4000/81}{100} = \frac{40}{81}$ .

5. Check: the center is at height  $\frac{10}{3}$  on the axis and its distance to a slant side is  $\frac{5(12 - \frac{10}{3})}{13} = \frac{10}{3}$ .

**Answer:**  $\frac{40}{81}$

42.

SAT CHALLENGE · CALCULATOR

A museum has a small solid model of a sculpture and a full-size copy; the two are similar solids. The sealant needed to coat the whole surface is proportional to surface area. The model needs 1.6 liters of sealant and the full-size copy needs 14.4 liters. The model has a volume of 2.5 cubic feet. What is the volume, in cubic feet, of the full-size copy?

(A) 22.5

**(B) 67.5 ✓**

(C) 202.5

(D) 7.5

1. Sealant is proportional to surface area, so the area ratio is  $\frac{14.4}{1.6} = 9 = k^2$ .
2. The linear scale factor is  $k = \sqrt{9} = 3$ .
3. Volumes scale by  $k^3 = 27$ :  $2.5 \cdot 27 = 67.5$ .
4. (Using the area ratio 9 directly would give 22.5, the wrong quantity.)

**Answer: (B)**

43.

SAT CHALLENGE · NO CALCULATOR NEEDED

Two aquariums are similar in shape. The smaller aquarium holds 250 liters and has 400 square decimeters of glass in total. The larger aquarium holds 432 liters. How many square decimeters of glass does the larger aquarium have in total? (Note that 1 liter = 1 cubic decimeter.)

1. Volumes scale by  $k^3$ :  $k^3 = \frac{432}{250} = \frac{216}{125}$ , so  $k = \sqrt[3]{\frac{216}{125}} = \frac{6}{5}$ .
2. Glass area scales by  $k^2 = \frac{36}{25}$ .
3.  $400 \cdot \frac{36}{25} = 576$  square decimeters.

**Answer: 576**

44.

SAT CHALLENGE · NO CALCULATOR NEEDED

A roof ornament is a solid pyramid whose base is a square of area 100 square inches. The apex is directly above the center of the base, and the volume of the pyramid is 400 cubic inches. What is the combined area, in square inches, of the four triangular lateral faces?

(A) 130 (B) 360

(C) 260 ✓ (D) 240

1. Base area 100 gives side  $s = 10$ . From  $V = \frac{1}{3}Bh$ :  $400 = \frac{1}{3}(100)h$ , so the vertical height is  $h = 12$ .
2. The height of a triangular face (the slant height) is the hypotenuse of a right triangle with legs  $h = 12$  and  $\frac{s}{2} = 5$ :  
 $\ell = \sqrt{144 + 25} = 13$ .
3. Each face has area  $\frac{1}{2}(10)(13) = 65$ . Four faces:  $4 \cdot 65 = 260$ .
4. (Option 240 uses the vertical height 12 as the face height; 130 is two faces; 360 adds the base.)

**Answer: (C)**